Name ____ Date ___________ Period _____

Continuous vs. Discrete

Algebra II Honors | Unit 1 | Day 3 | TEKS A2.2.A, A2.7.I

Objective: I can classify a domain as continuous or discrete, write each with the notation that fits, recognize step behavior, and judge when discreteness is a property of the quantity versus a modeling choice.

Vocabulary

Continuous — every value between any two points belongs to the domain. **Discrete** — the domain is ___________________.

Step function — a _____________ domain paired with a _____________ range. The output jumps.

Sequence — a function whose domain is the __________________. Always discrete.

Part 1 — Classify and Justify

Write C or D, then give the domain in correct notation and one sentence of justification.

Situation	C/D	Domain	Why
Distance a car travels in 3 hours	_____	____________	________________
Number of students on 6 buses	_____	____________	________________
Mass of water in a beaker	_____	____________	________________
Term number in a sequence	_____	____________	________________

Part 2 — One Equation, Two Domains

A cube-shaped box of side x has volume $V(x) = x^3$.

x	1	2	3	4
V(x)	1	8	27	64

Any size box: domain ___________________ range ___________________ _______________

Built from 1-inch blocks: domain _____________________ range _____________________ _______________

The algebra did not change. What did? __________________________________

Part 3 — Step Behavior

A parking garage charges $3 for each hour or any part of an hour, up to 4 hours. Park for 1.2 hours and you pay for 2.

t (hours)	0.5	1.0	1.2	2.0	2.7	3.5
Cost						

Domain (time parked): ___________________ _______________ Range (cost): _______________________ _______________

This is the case the on-level notes do not cover: a domain and range that disagree. Explain how that happens.

____

____

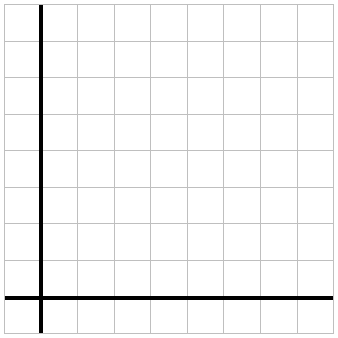
Time varies smoothly, but the _________________ at each whole hour instead of rising gradually.

Why must t = 1.0 cost $3 and not $6? ____

Part 4 — Graph Both Kinds

Each square is 1 unit. Use open and closed dots carefully.

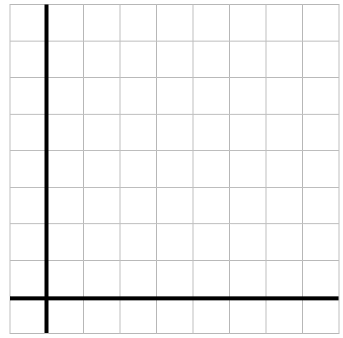
A. Discrete. A tutor charges $2 per session, up to 4 sessions.



Domain ___________________

Range ___________________

B. Continuous. A candle burns 2 cm per hour from 8 cm.



Rule h = _______________

Domain _______________ Range _______________

Part 5 — Going Further

1. Is money continuous or discrete? Argue both sides, then commit to one.

____

____

Discrete in practice _____________________________, often modeled as continuous because the steps are tiny.

2. Give a situation with a discrete domain and a continuous range, or argue it is impossible.

____

____

Impossible — ____

Exit Ticket

Postage costs $0.60 for the first ounce and $0.24 for each additional ounce or part of an ounce, up to 5 ounces.

Domain (weight) ______________________ Range (cost) _________________________________

Classify the domain and the range, and explain why they differ: ____